MEASUREMENTS IN THE CONTINUUM SOIL-PLANT-ATMOSPHERE

Víctor Blanco Montoya

IRTA - Efficient Use of Water in Agriculture Program

February 5th and 6th, 2024

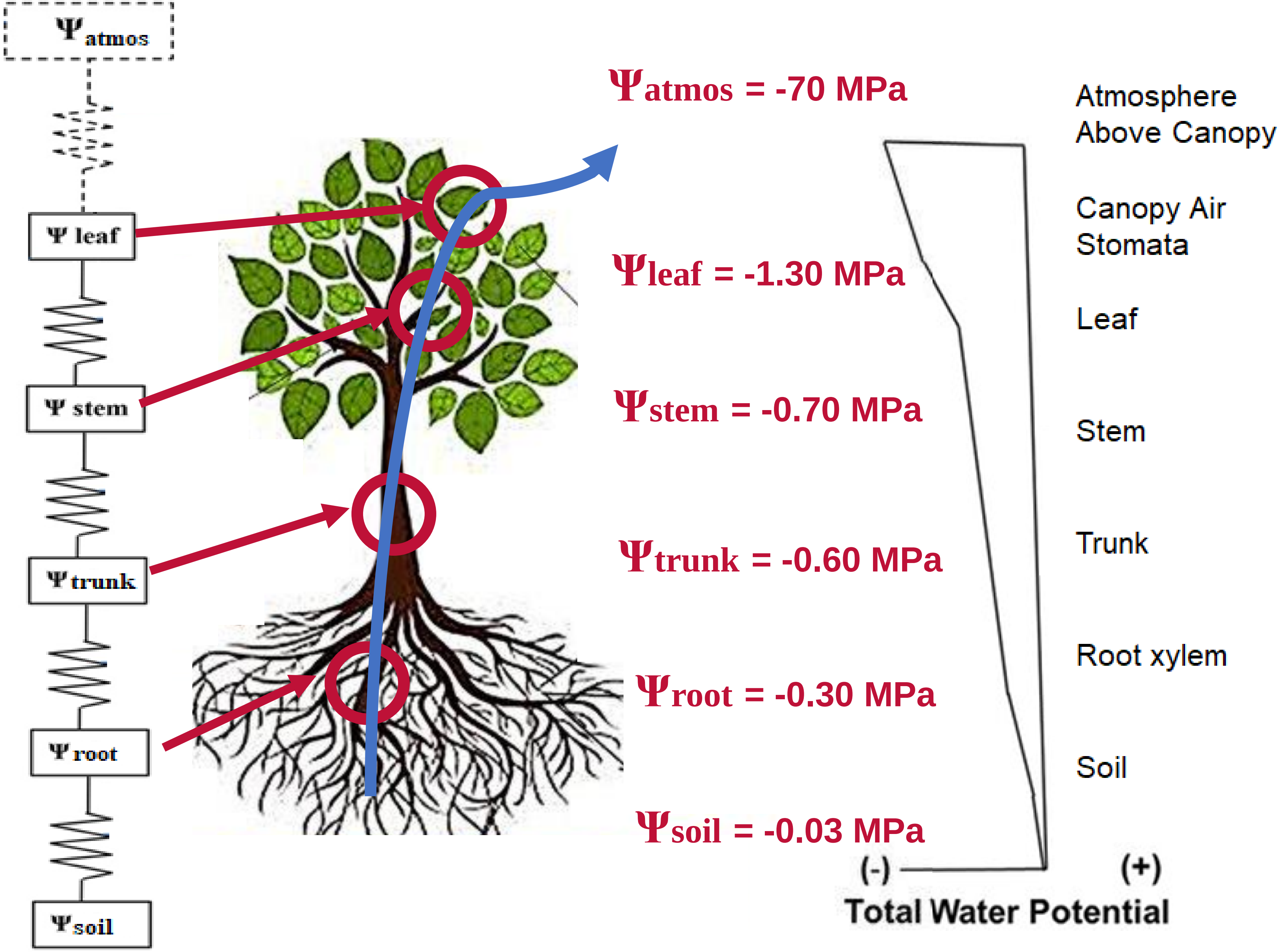
FLORAPULSE WORKSHOP

SOIL–PLANT–ATMOSPHERE CONTINUUM

The soil-plant-atmosphere continuum (SPAC) is the pathway for water moving from soil through plants to the atmosphere.

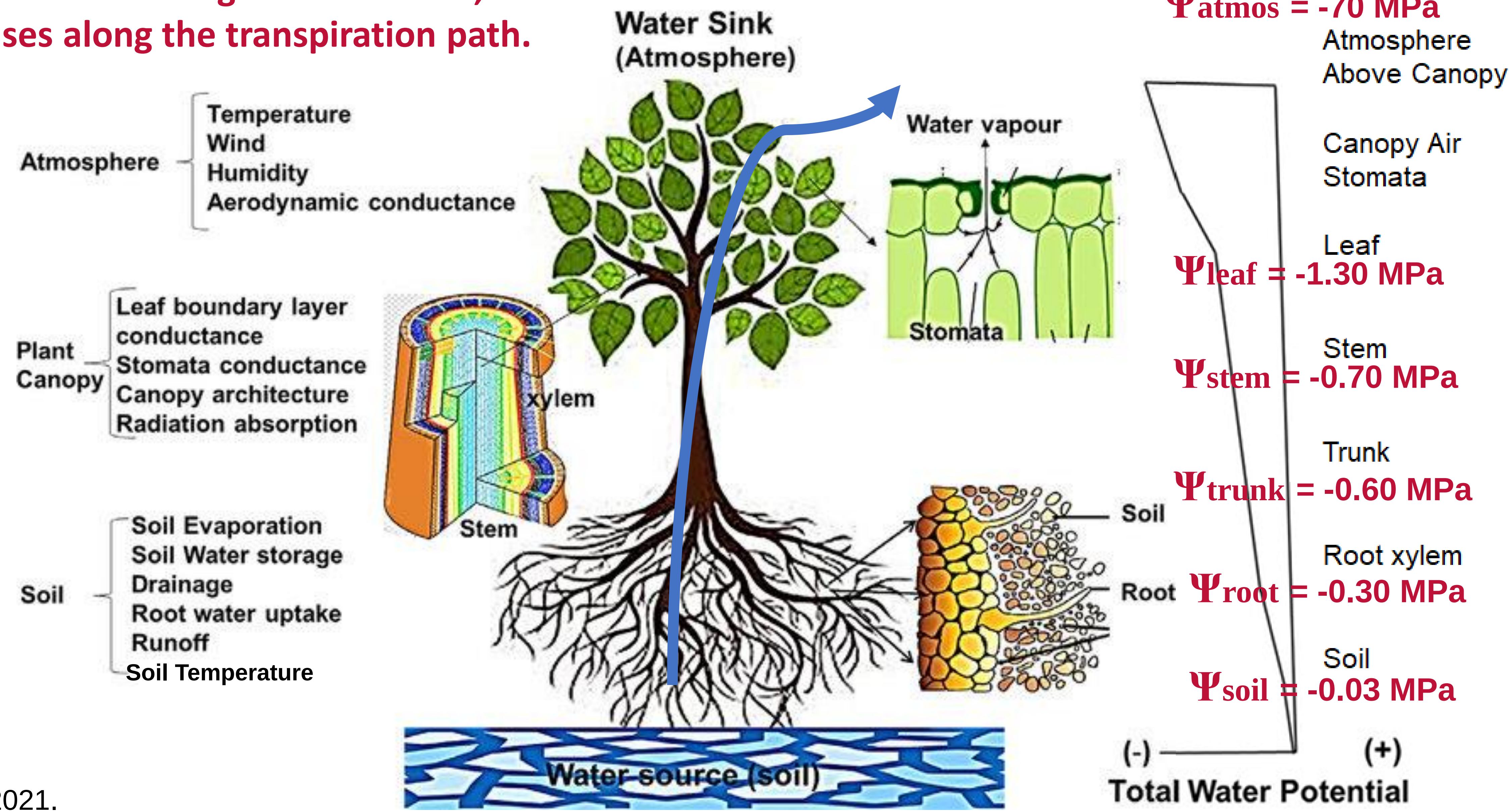
SPAC lies in water movement through plants from areas of high (less negative) to low (more negative) water potential.

This potential gradient provides the driving force for water transport from the soil to the atmosphere.



SOIL–PLANT–ATMOSPHERE CONTINUUM

Water potential is the highest in the soil, and decreases along the transpiration path.



Rekwar et al. 2021.
Ecological aspects of the soil-water-plant-atmosphere system

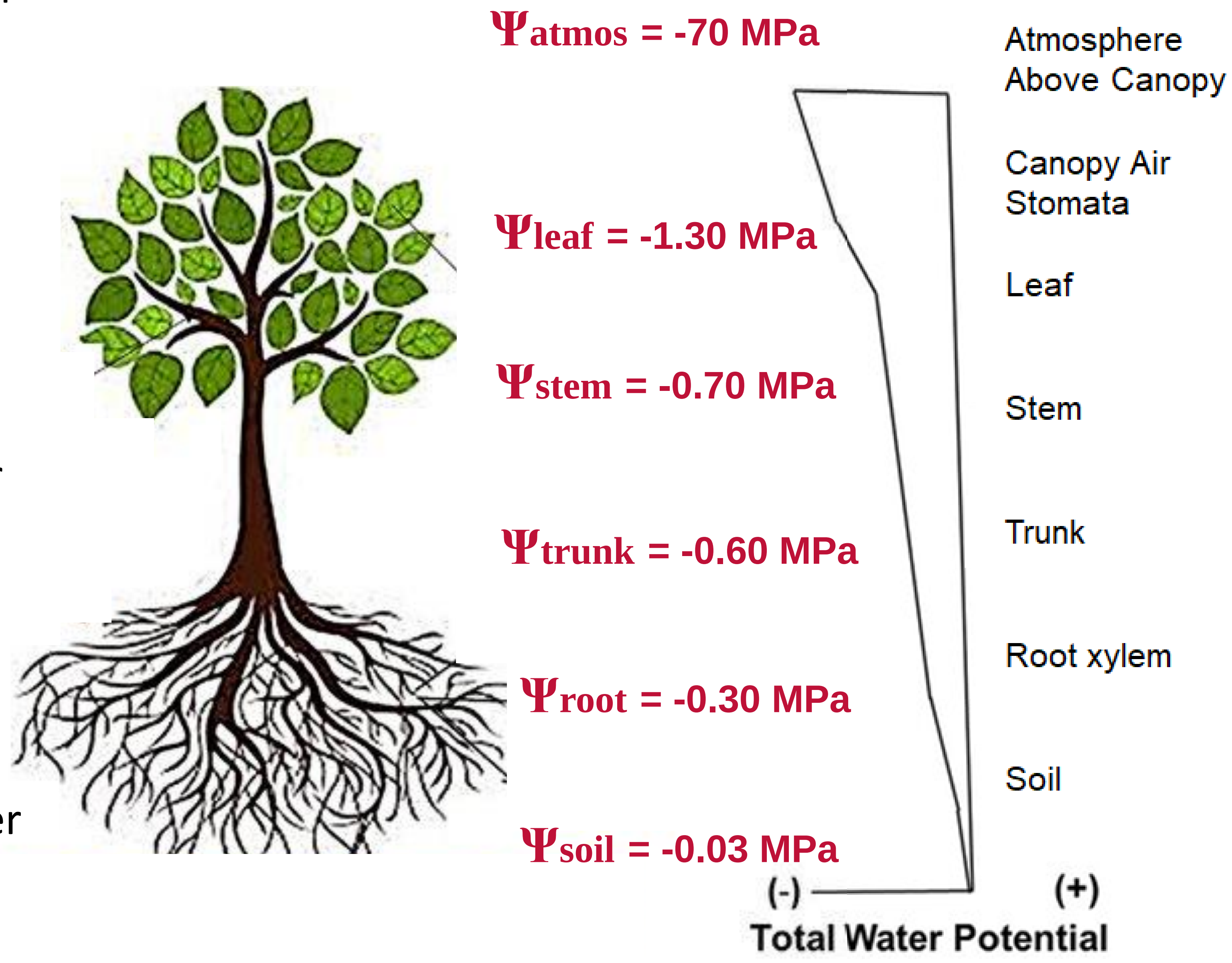
SOIL–PLANT–ATMOSPHERE CONTINUUM

Traditionally tree water status has been estimated from measurement of soil and atmosphere water status and/or discrete measurements of plant water status.

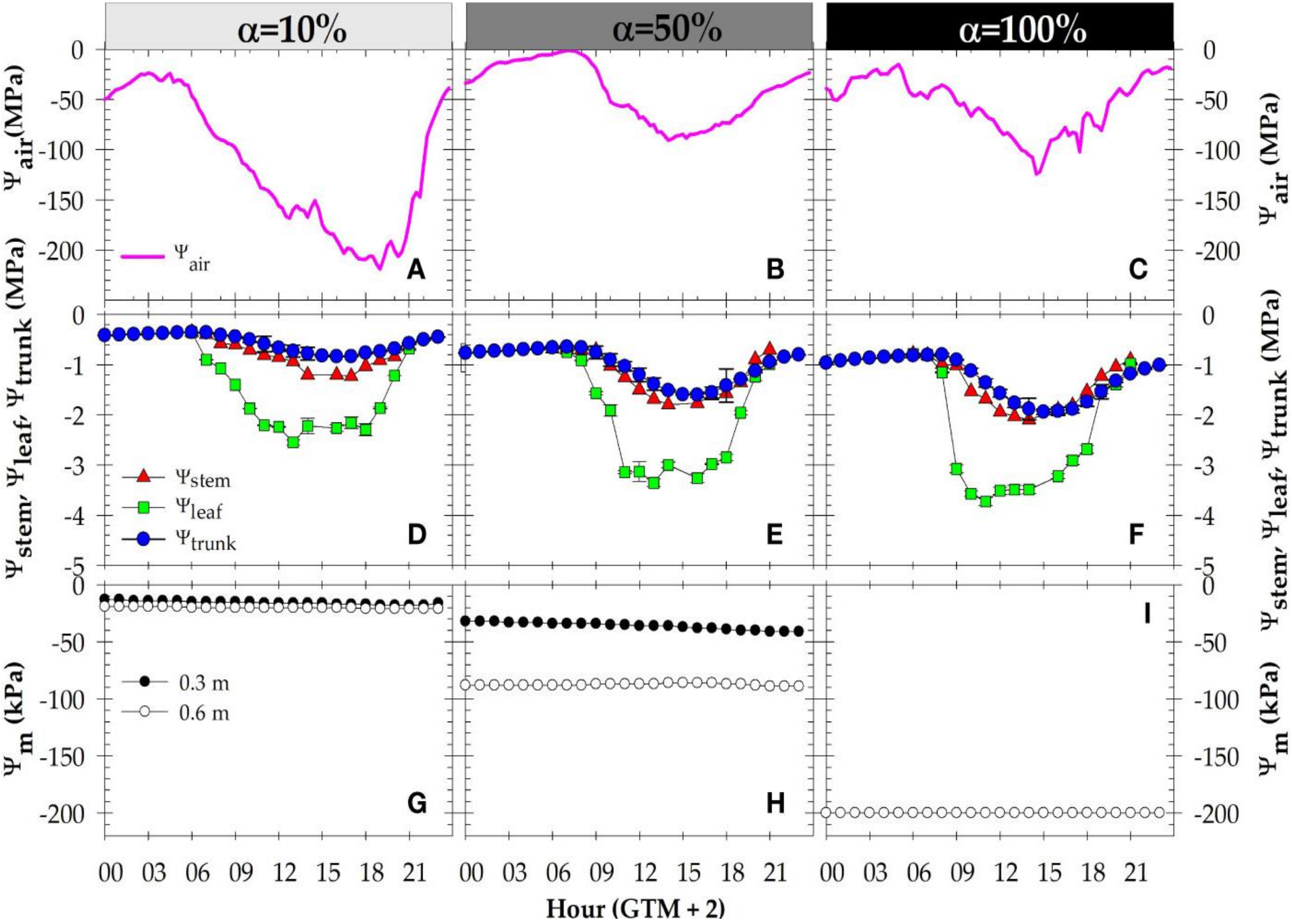
Ψstem & Ψleaf

Stem water potential represents the whole canopy and integrates soil water availability and atmospheric water demand. It is related to processes such as vegetative growth and fruit development.

Now we can monitor a new plant water status indicator **Ψtrunk**



SOIL–PLANT–ATMOSPHERE CONTINUUM



Daily evolution

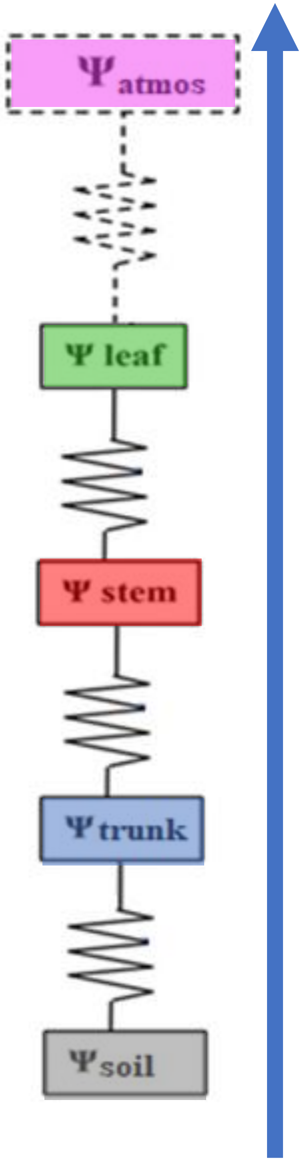
Ψatmosphere

Ψleaf

Ψstem

Ψtrunk

Ψsoil



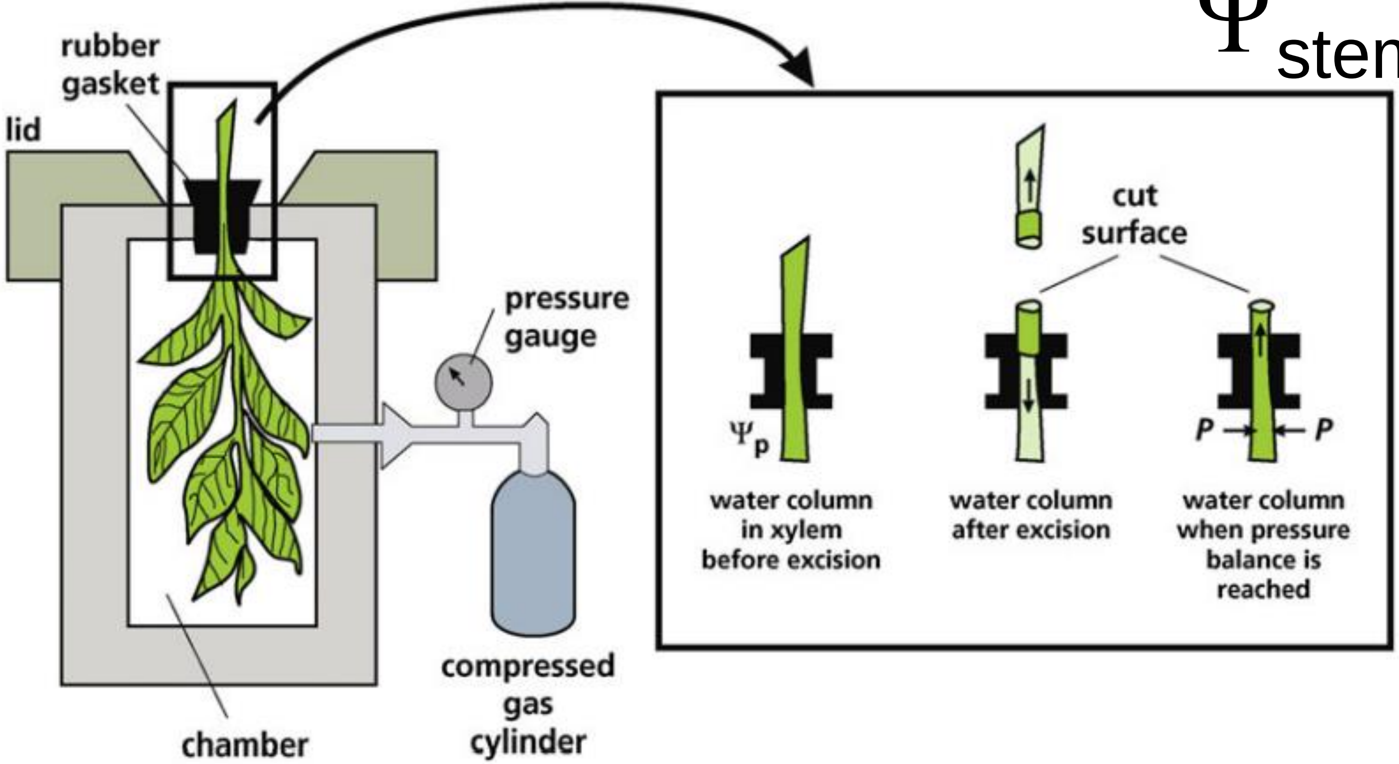
Conesa et al. 2023

Assessment of trunk microtensiometer as a novel biosensor to continuously monitor plant water status in nectarine trees

STEM WATER POTENTIAL | **LEAF WATER POTENTIAL**

Pressure Chamber – Scholander Pressure Bomb

Ψ_{stem} MPa



Pressure

The pressure chamber measures the water potential by applying pressurized gas to a chamber where the excised leaf is placed with the petiole extending outside the sealed chamber.

Lambers and Oliveira, 2019. Plant water relations
Irrigation Scheduling using Stem Water Potential (SWP) Measurements. UCDAVIS. Fruit & nut research & information.

PRESSURE CHAMBER

MIDDAY STEM WATER POTENTIAL

Pressure Chamber – Scholander Pressure Bomb

Wait for 1 to 2 hours Equilibrium between leaves and stem

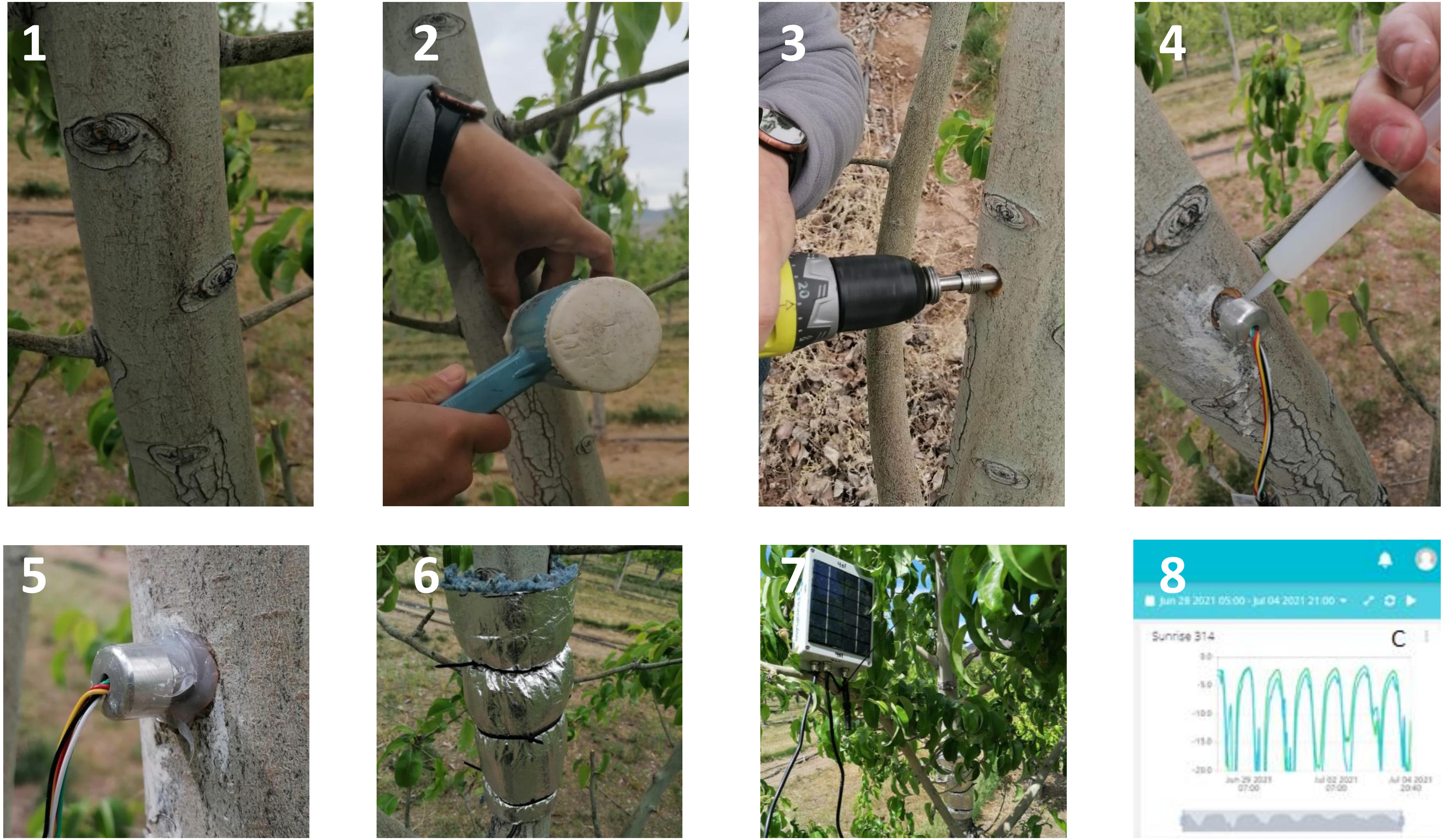
Robust, direct, and reliable indicator for fruit trees and vines.

It is a temporally discrete, labor-demanding, destructive measurement.

MICROTENSIOMETERS

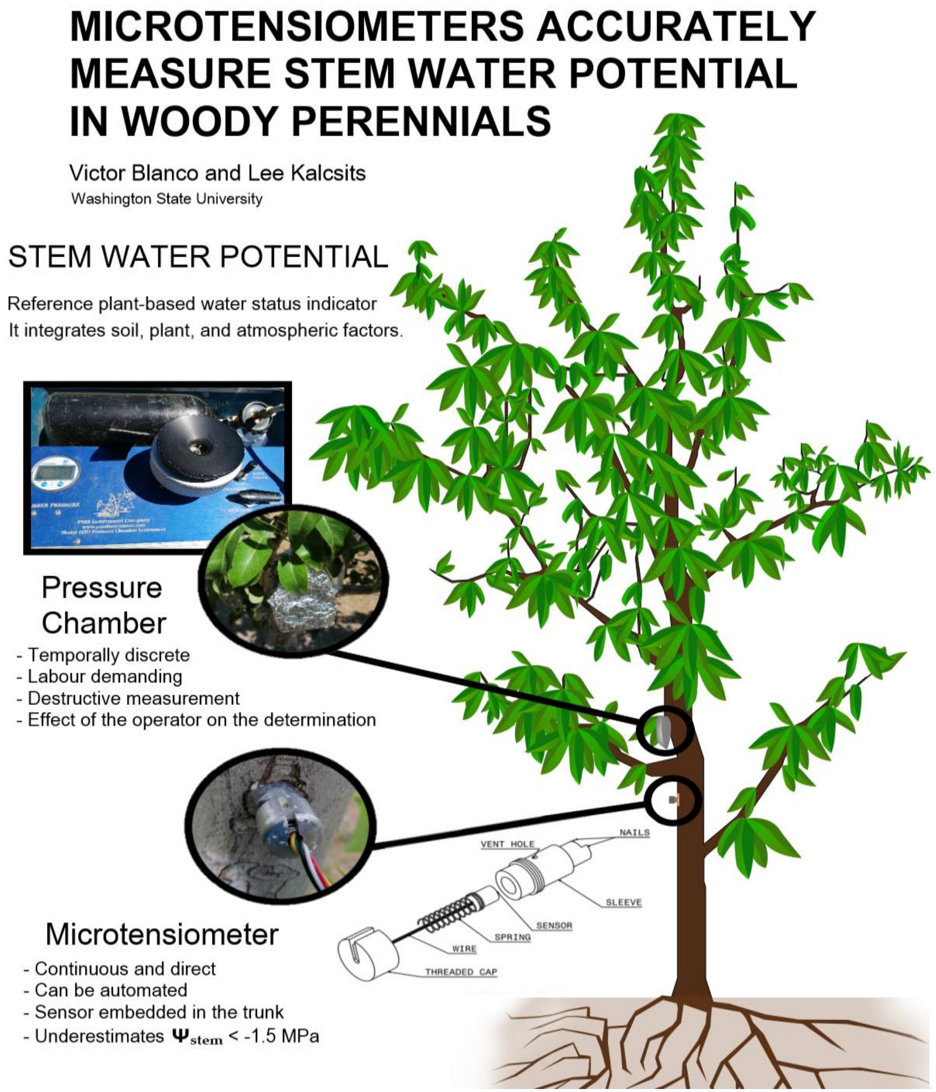
New sensor – Continuous monitoring trunk water potential

FLORAPULSE - MICROTENSIOMETER INSTALLATION



MICROTENSIOMETERS

TRUNK WATER POTENTIAL



Ψ_{trunk} MPa

Objective: Compare trunk water potential values acquired from **FloraPulse microtensiometers** with those values of stem water potential measured with the pressure chamber and other water status indicators.

Blanco and Kalcsits, 2021. Microtensiometers

RECENTLY PUBLISHED PAPERS

From the journal:
Lab on a Chip

A microtensiometer capable of measuring water potentials below −10 MPa†

Vinay Pagay,‡[a] Michael Santiago,[b] David A. Sessoms,[c] Erik J. Huber,[b] Olivier Vincent,[c] Amit Pharkya,[c] Thomas N. Corso,[d] Alan N. Lakso[e] and Abraham D. Stroock*[cf]

Contents lists available at ScienceDirect
Agricultural Water Management
journal homepage: www.elsevier.com/locate/agwat

Long-term validation of continuous measurements of trunk water potential and trunk diameter indicate different diurnal patterns for pear under water limitations

Victor Blanco[a,b], Lee Kalcsits[a,b,*]

plants MDPI

Article

Trunk Water Potential Measured with Microtensiometers for Managing Water Stress in "Gala" Apple Trees

Luis Gonzalez Nieto[1,*], Annika Huber[2], Rui Gao[2], Erica Casagrande Biasuz[1], Lailiang Cheng[1], Abraham D. Stroock[2,3], Alan N. Lakso[1,3] and Terence L. Robinson[1]

TECHNICAL BRIEF

Irrigation Science (2022) 40:45–54
https://doi.org/10.1007/s00271-021-00758-8

Evaluating a novel microtensiometer for continuous trunk water potential measurements in field-grown irrigated grapevines

Vinay Pagay[1]

horticulturae

Article

Monitoring Stem Water Potential with an Embedded Microtensiometer to Inform Irrigation Scheduling in Fruit Crops

Alan N. Lakso[1,2,*], Michael Santiago[1] and Abraham D. Stroock[3]

plants

Article

Microtensiometers Accurately Measure Stem Water Potential in Woody Perennials

Victor Blanco[1,2] and Lee Kalcsits[1,2,*]

frontiers | Frontiers in Plant Science

Assessment of trunk microtensiometer as a novel biosensor to continuously monitor plant water status in nectarine trees

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WASHINGTON STATE UNIVERSITY

MICROTENSIOMETERS.
CONTINUOUS MONITORING OF TREE WATER STATUS

Victor Blanco and Lee Kalcsits
Washington State University

Experiments:

1. Validation

1.1 Daily evolution. Blanco and Kalcsits, 2021. *Plants*
https://doi.org/10.3390/plants10122780

1.2 Seasonal evolution. Blanco and Kalcsits, 2023. *Agric. Water Manag.*
https://doi.org/10.1016/j.agwat.2023.108257

Materials & Methods

EXPERIMENTAL CONDITIONS

Pear trees **'D'Anjou'/'OHxF.87'** 2007

Central Washington – Semiarid climate

Central leader system

Tree Density: 833 tree ha^{-1}

Drip irrigation – 1 Line 5 em tree^{-1}. 2 L h^{-1}

EVALUATION - Validation

Microtensiometers vs Pressure Chamber

Experiment 1:

4 representative days – Season 2021

Experiment 2:

Season 2021 and season 2022

Materials & Methods

IRRIGATION TREATMENTS

	April	May	June	July	August	Sept	Oct	Nov
CTL	**100 % ETc**			**100 % ETc**				
DI	**100 % ETc**			**50 % ETc**				

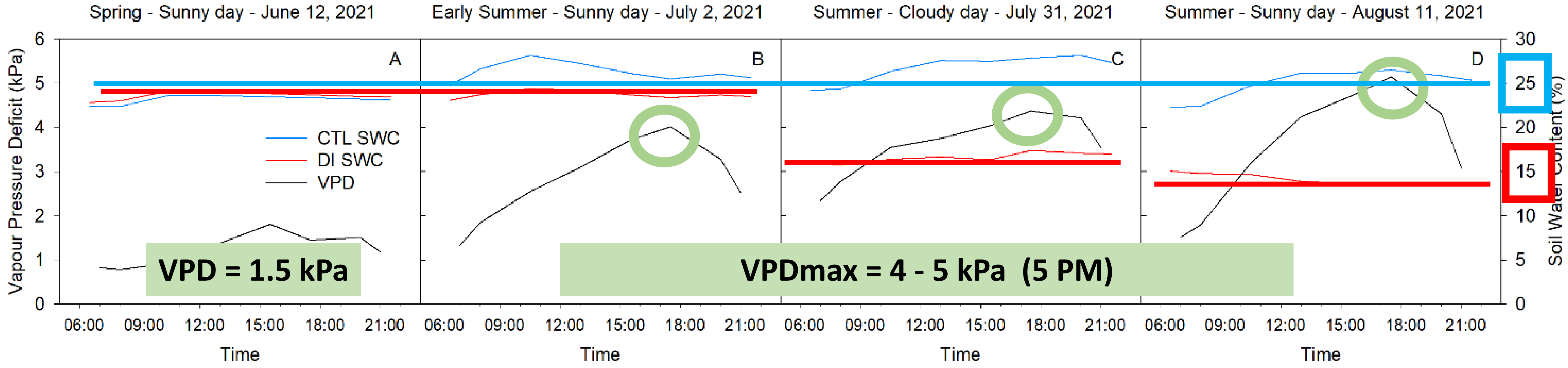
Bloom : April   Second Stage Fruit Growth: June - July   Harvest: September

EXPERIMENTAL DESIGN - MEASUREMENTS

3 replicates per treatment – 2 trees per replicate = 6 trees per treatment Randomly distributed

Soil and Plant Water Status and Environmental Demand

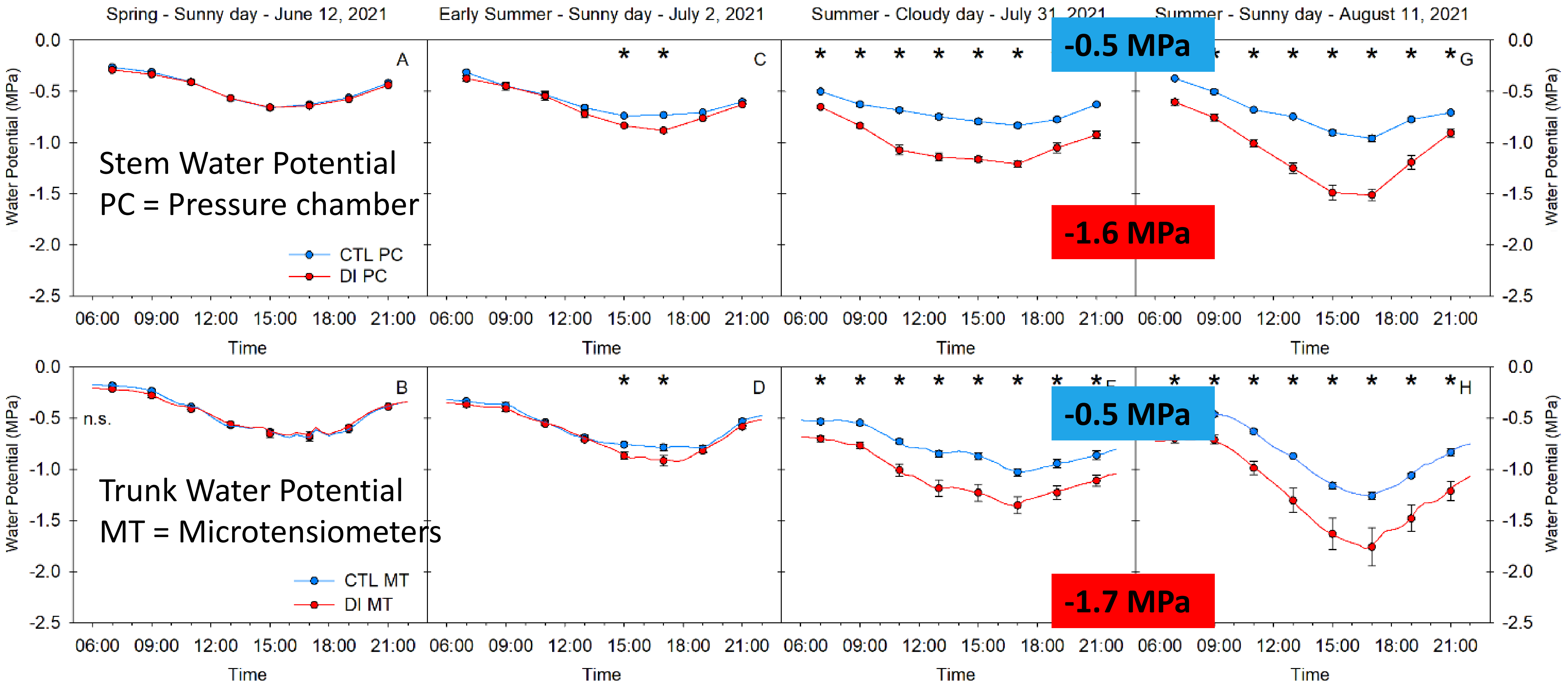
ENVIRONMENTAL CONDITION AND SOIL WATER CONTENT



	Sunny day Low atmospheric demand June 12, 2021	Sunny day High atmospheric demand July 2, 2021	Cloudy day High atmospheric demand July 31, 2021	Sunny day High atmospheric demand August 11, 2021
CTL	**100 % ETc**	**100 % ETc**		
DI	**100 % ETc**	**50 % ETc**		

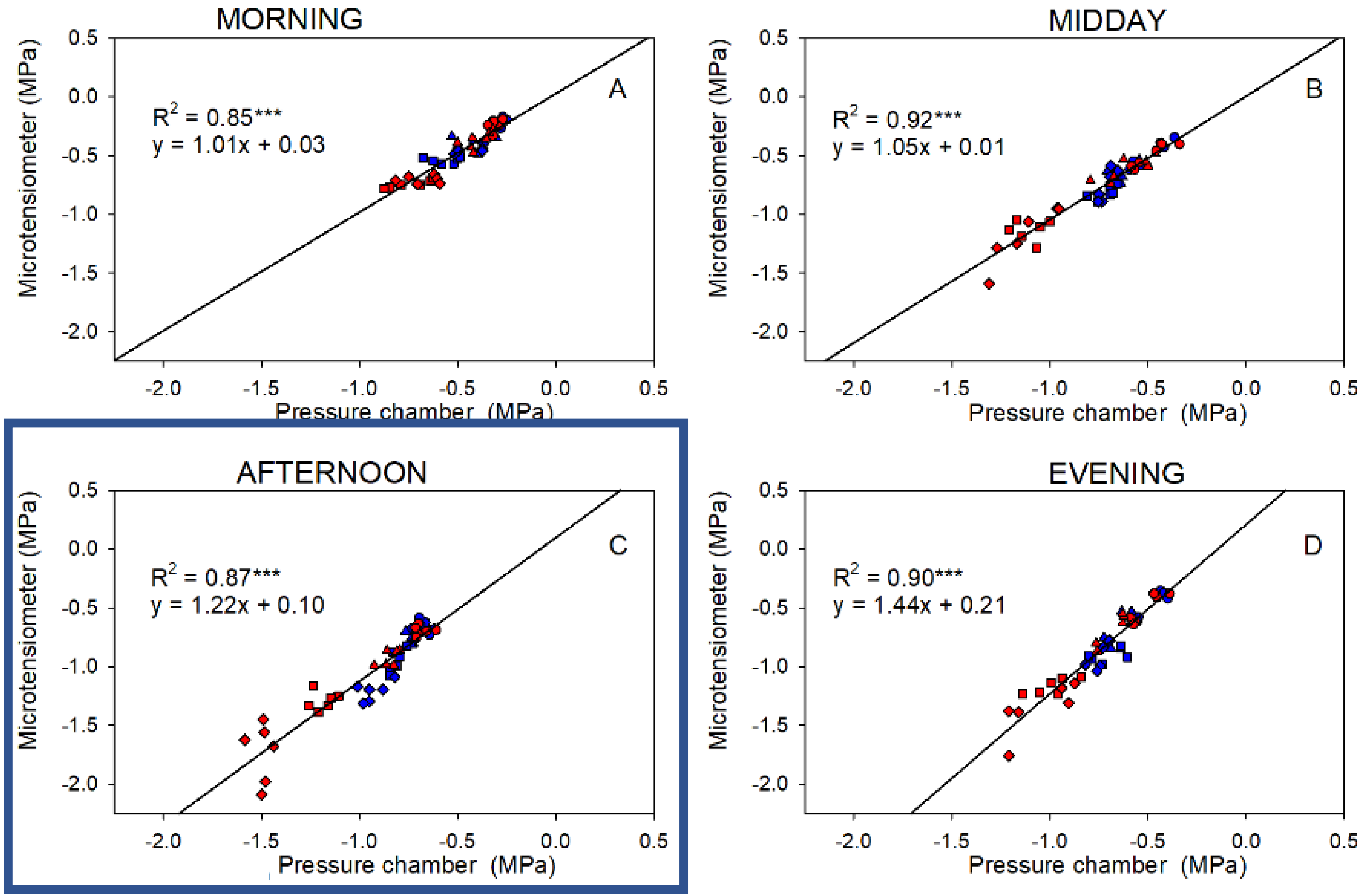
Results – Daily Evolution

STEM & TRUNK WATER POTENTIAL



Results – Daily Evolution

Parts of the day – Different response



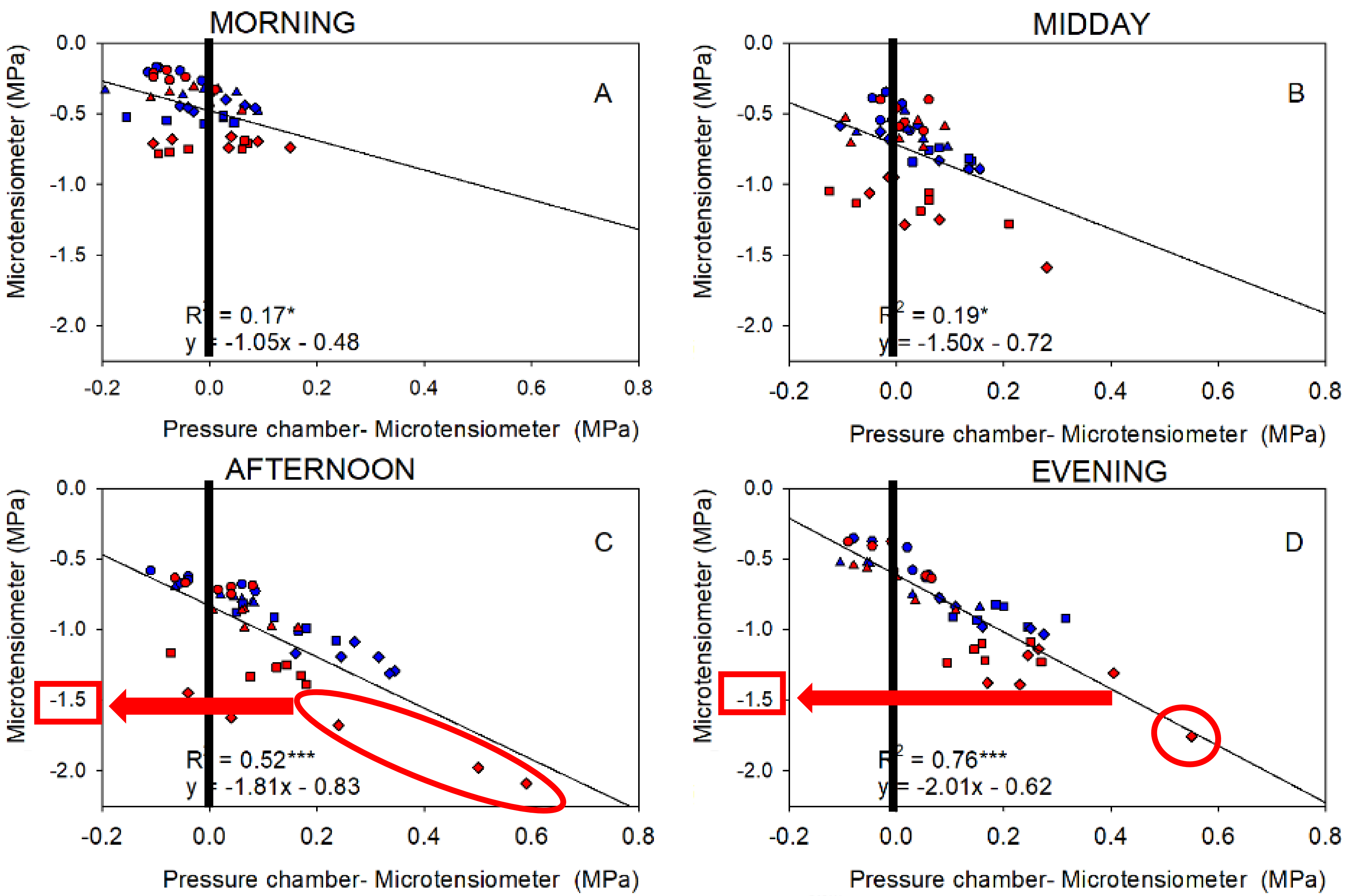
Low variability during the morning.

High variability during the afternoon when the highest VPD is recorded.

Results – Daily Evolution

Parts of the day – Different response

Ψtrunk *vs.* (Ψstem – Ψtrunk)

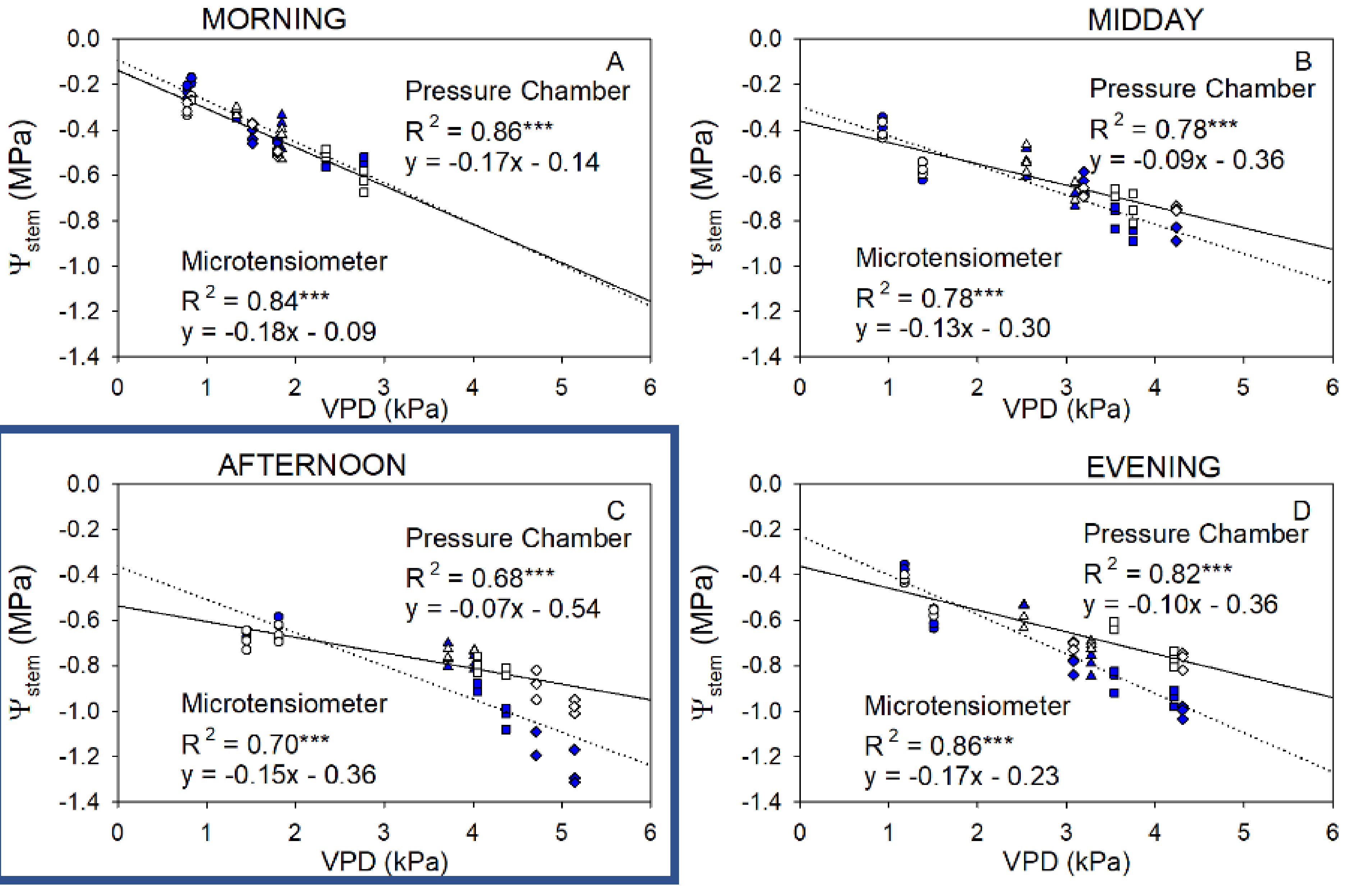


Ψtrunk was less negative than Ψstem during the morning, similar during the midday but more negative during the afternoon and evening.

Particularly for Ψtrunk values < -1.5 MPa

Results – Daily Evolution

Relationship between VPD and stem water potential (CTL)



Stem and Trunk water potentials were strongly related to VPD.

For VPD values:

0 – 2.5 kPa

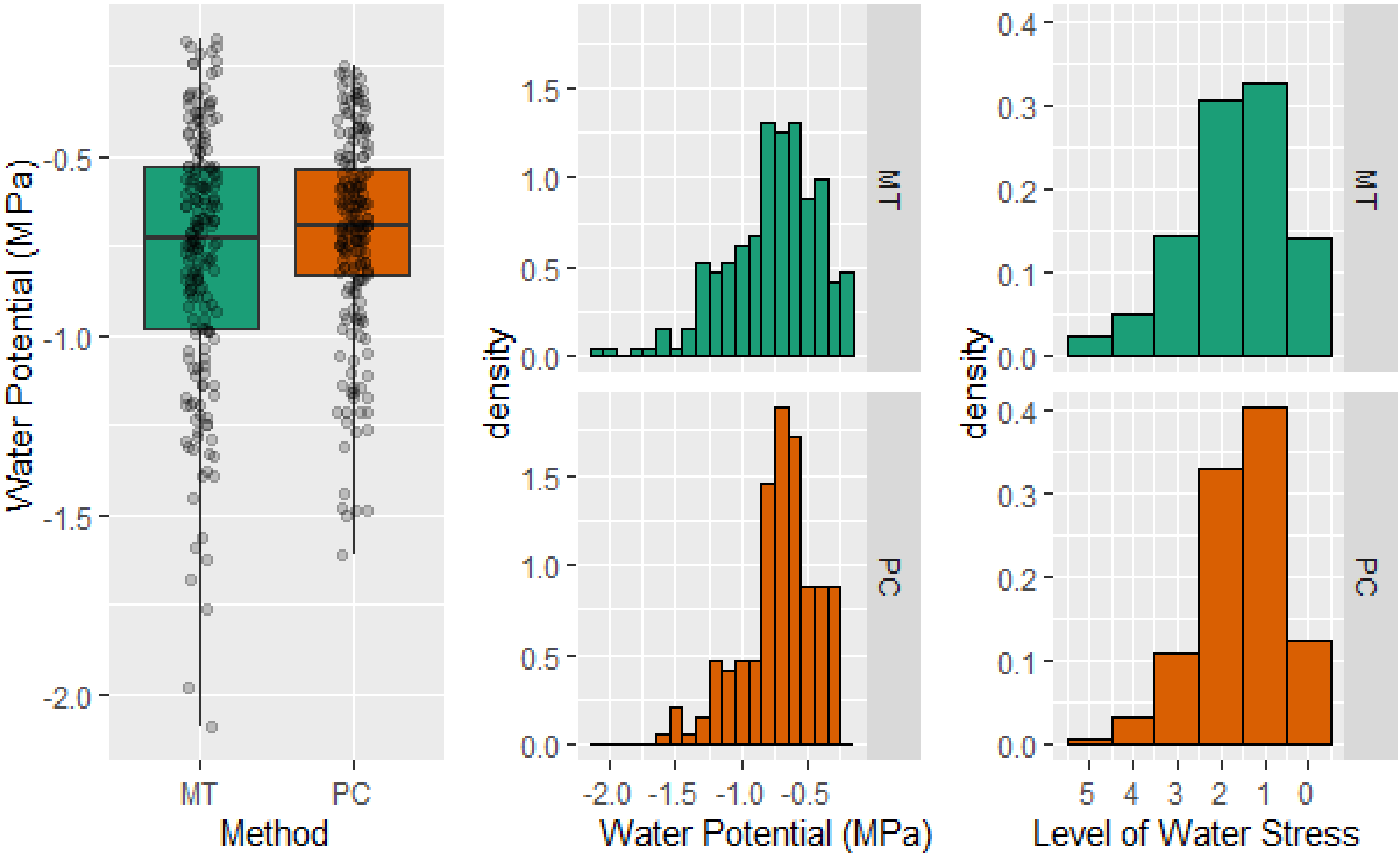
Ψtrunk > Ψstem

2.5 – 6 kPa

Ψtrunk < Ψstem

Results – Daily Evolution

Ψstem DENSITY AND DISTRIBUTION



Ψtrunk had greater variability

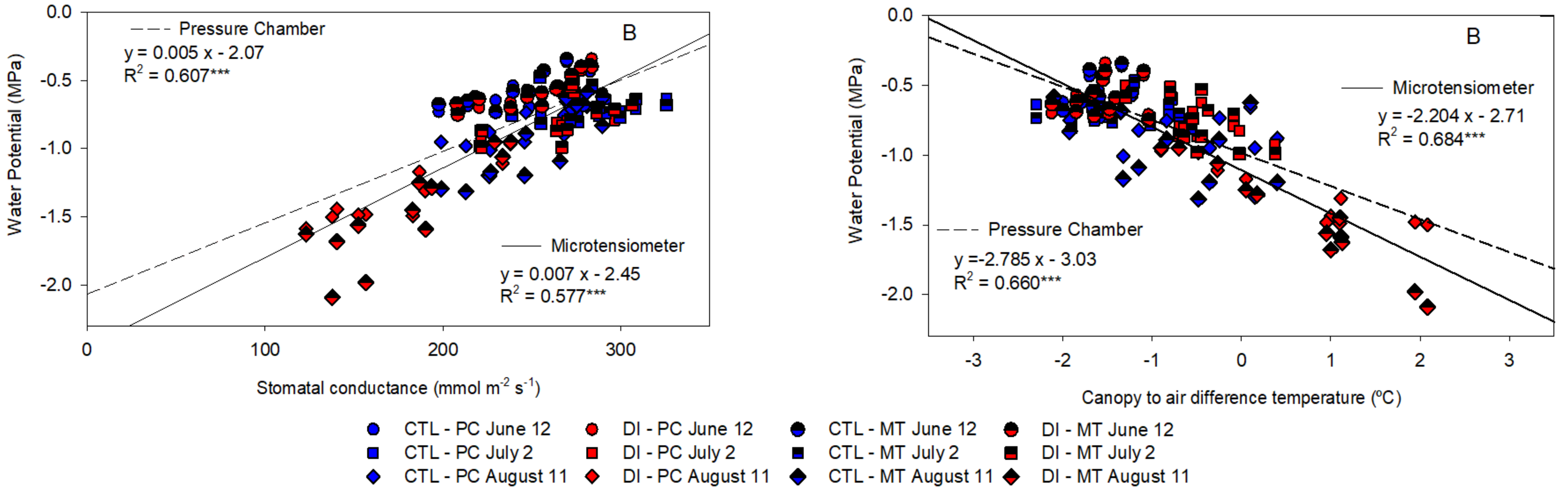
MT (Ψtrunk) had higher number of values < -1.5 MPa

Similar results when we consider ranges of water stress (levels).

Results – Daily Evolution

Ψstem *vs. Gs* Ψtrunk *vs. Gs*

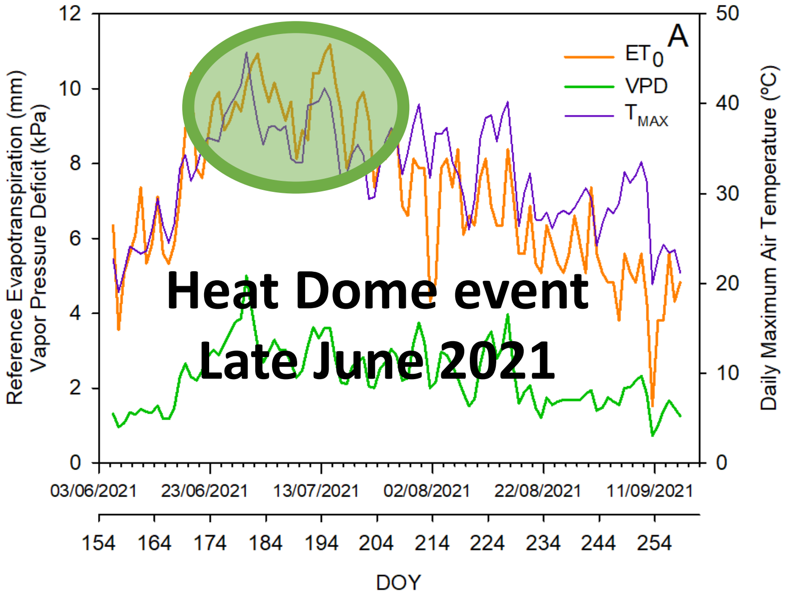
Ψstem *vs.* Canopy to Air Temp.
Ψtrunk *vs.* Canopy to Air Temp.

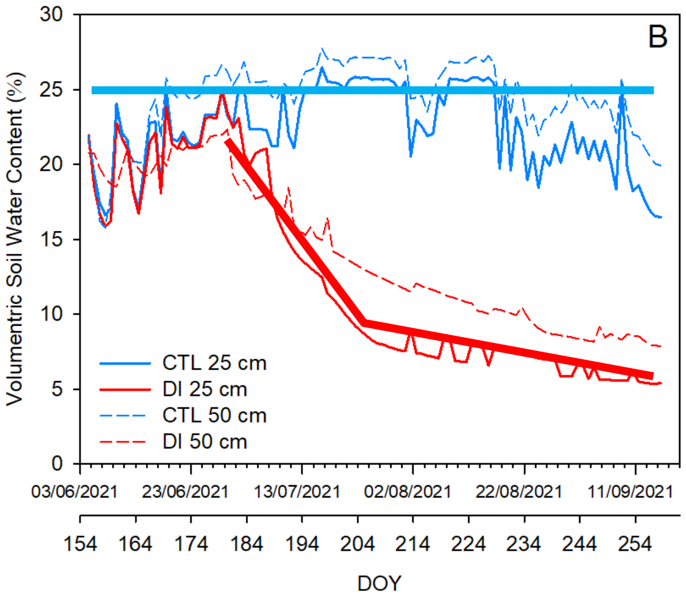


Stem water potential (PC) and trunk water potential (MT) were similarly related with the stomatal conductance and the canopy temperature.

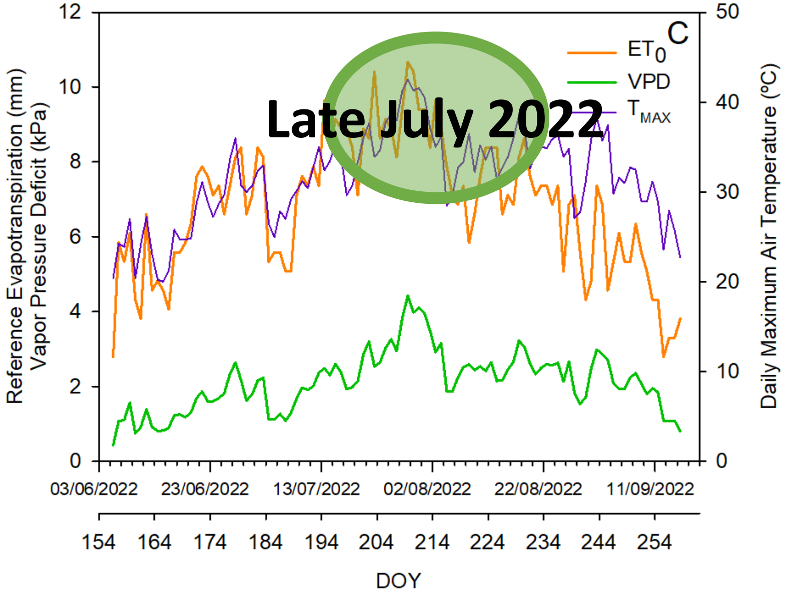
Results – Seasonal Evolution

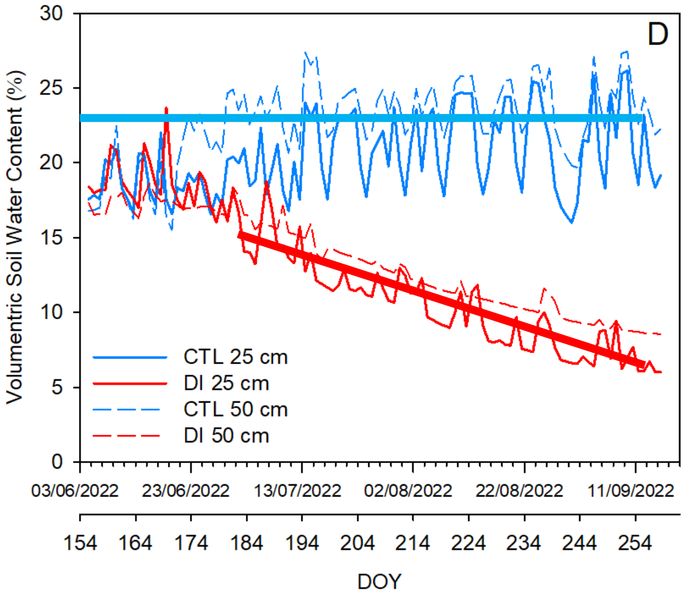
ENVIRONMENTAL CONDITION AND SOIL WATER CONTENT





Season 2021
Heatwave
Fast decrease in soil water content

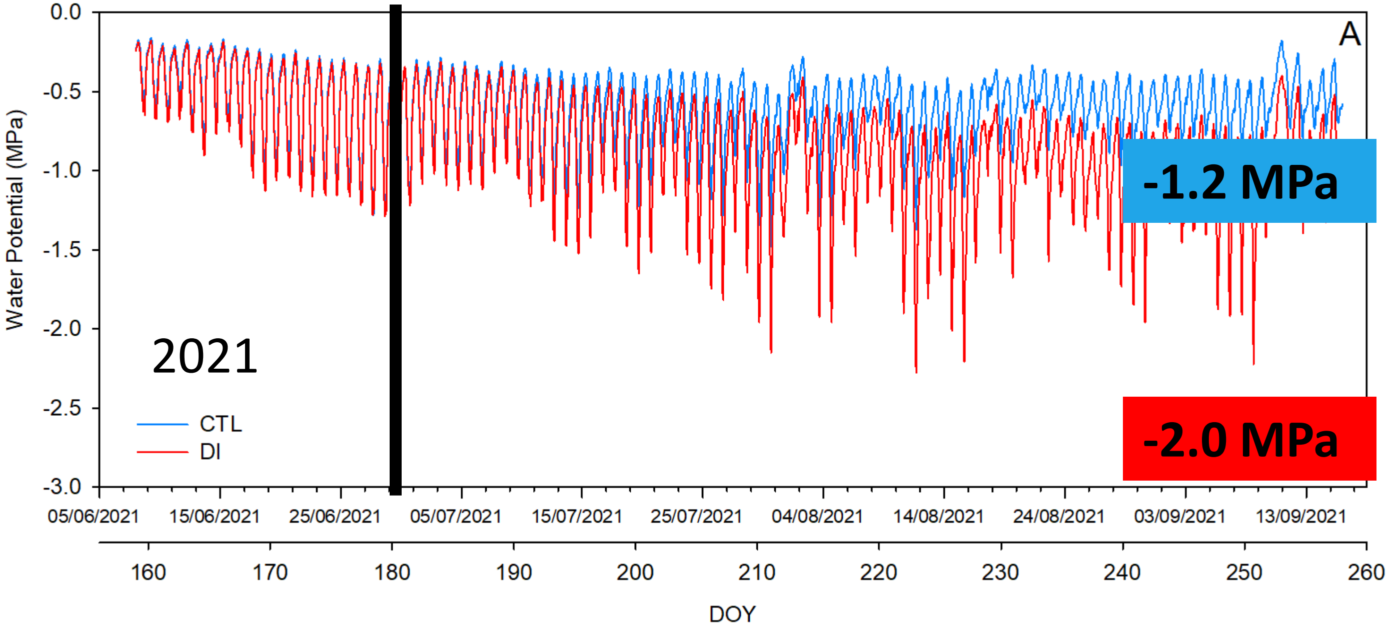




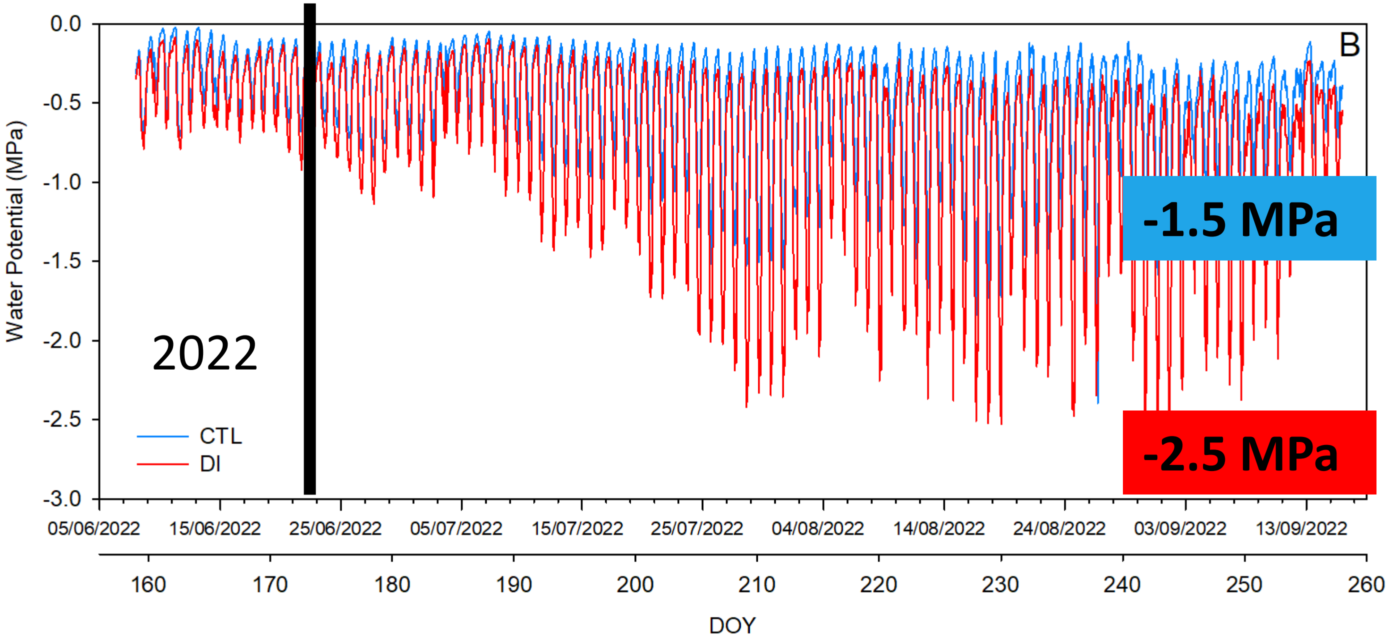
Season 2022
Slow decrease in soil water content

Results – Seasonal Evolution

TRUNK WATER POTENTIAL - MICROTENSIOMETERS



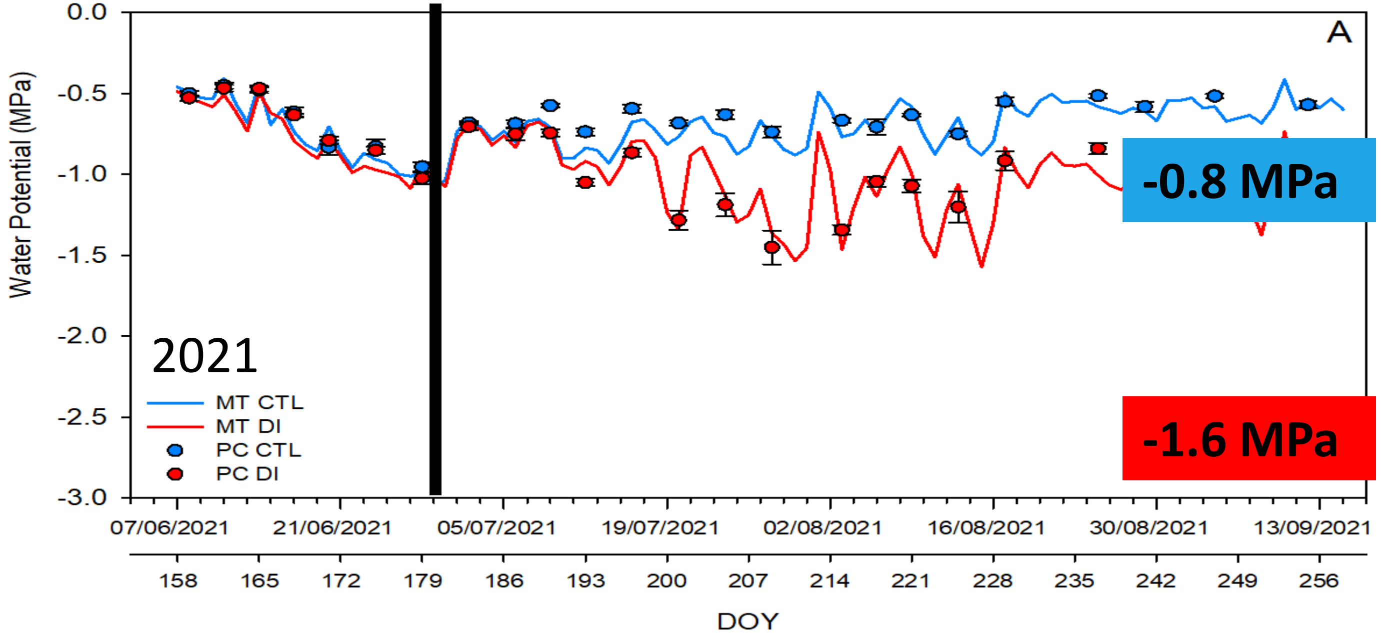
Both years before the irrigation change, the microtensiometers recorded similar results for both treatments.

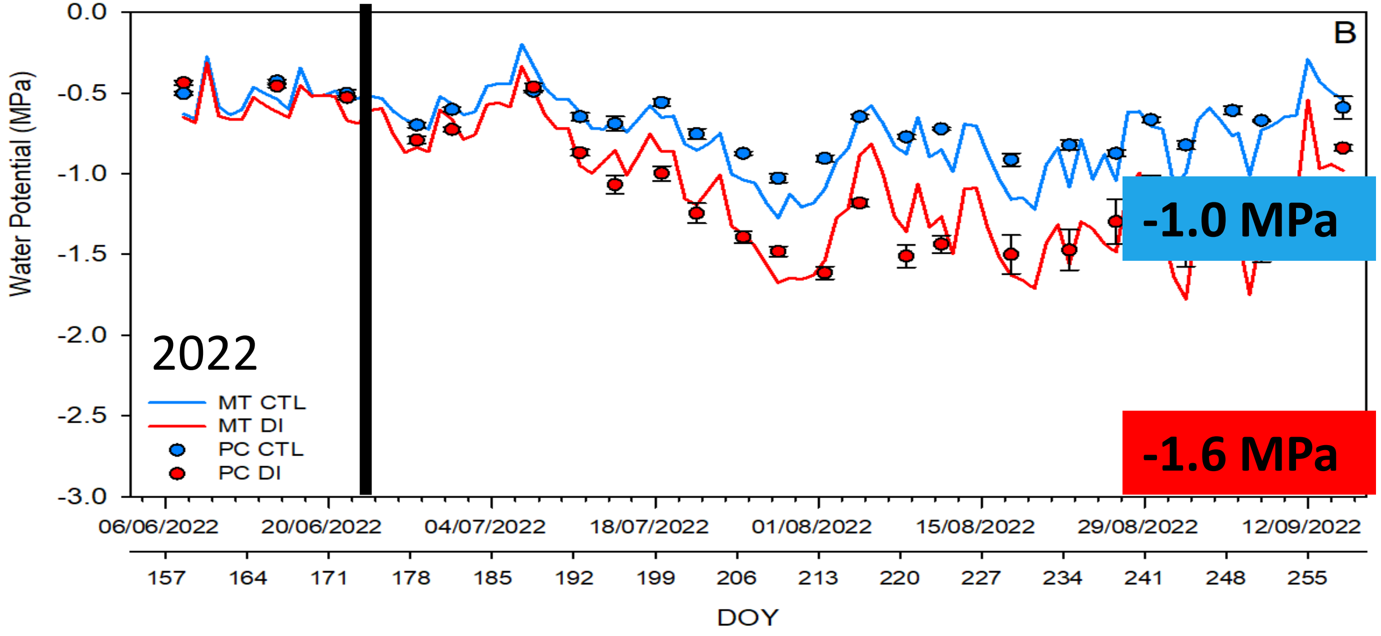


Daily minimum trunk water potential values were more negative in 2021 than in 2022.

Results – Seasonal Evolution

TRUNK WATER POTENTIAL - MICROTENSIOMETERS





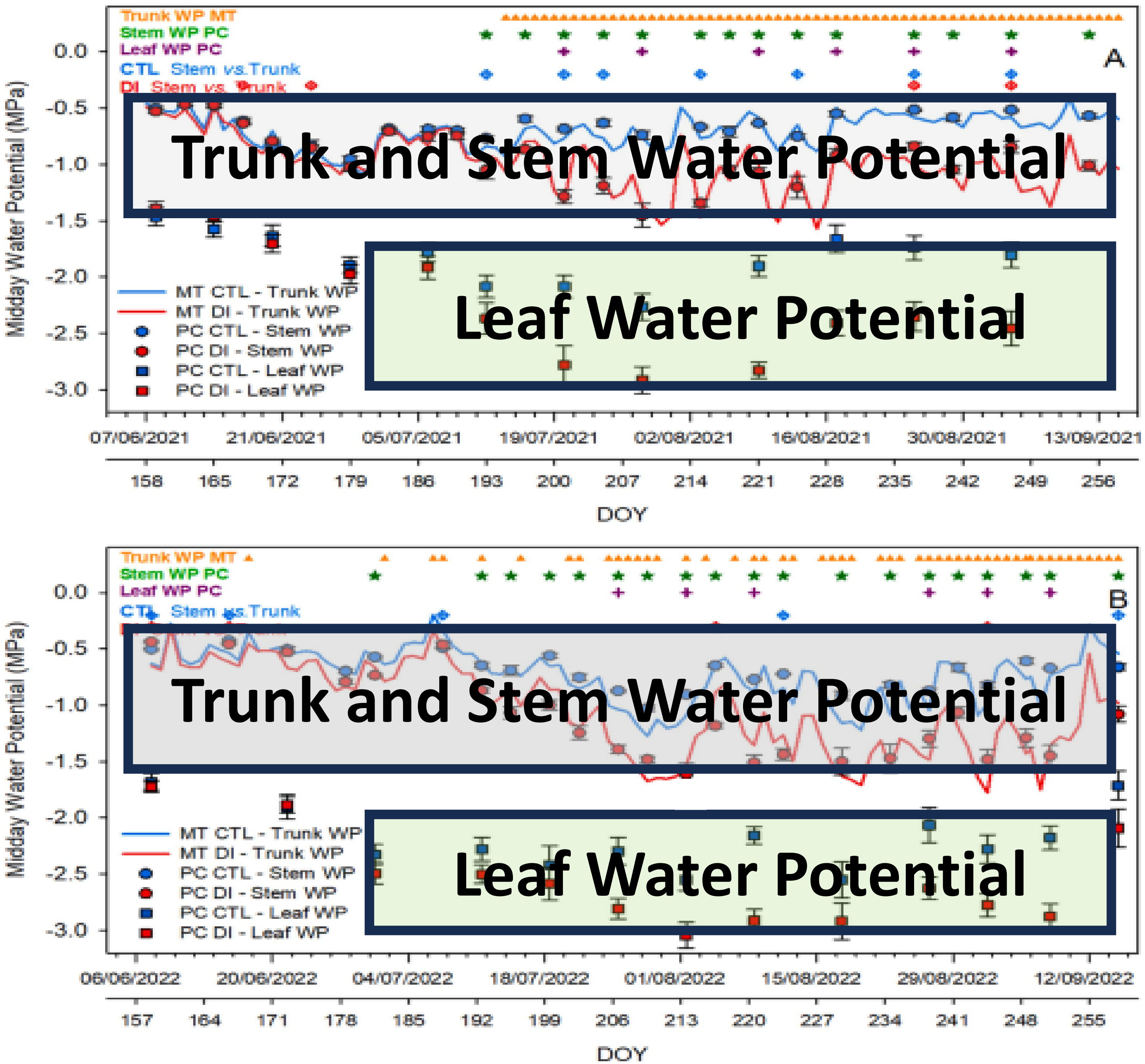
PC - MIDDAY STEM WATER POTENTIAL

MT - MIDDAY TRUNK WATER POTENTIAL

At midday Ψstem and Ψtrunk had similar results.

Tree water status was clearly affected by the irrigation treatment imposed.

Results – Seasonal Evolution



TRUNK WATER POTENTIAL
STEM WATER POTENTIAL
LEAF WATER POTENTIAL

Ψtrunk and Ψstem had similar values and recognized the irrigation treatments imposed.

Ψleaf was less sensitive than Ψtrunk and Ψstem detecting water deficit.

Results – Seasonal Evolution

DENDROMETERS

Sensor that measures small fluctuations in trunk or fruit diameter resulting from variation in sap flow.

Trunk and fruit dendrometers can be used to detect and quantify water stress to improve irrigation scheduling in fruit trees.

Trunk Dendrometer

Trunk dendrometers are devices that are attached to the trunk of a tree to measure daily variation in trunk diameter.

Results – Seasonal Evolution

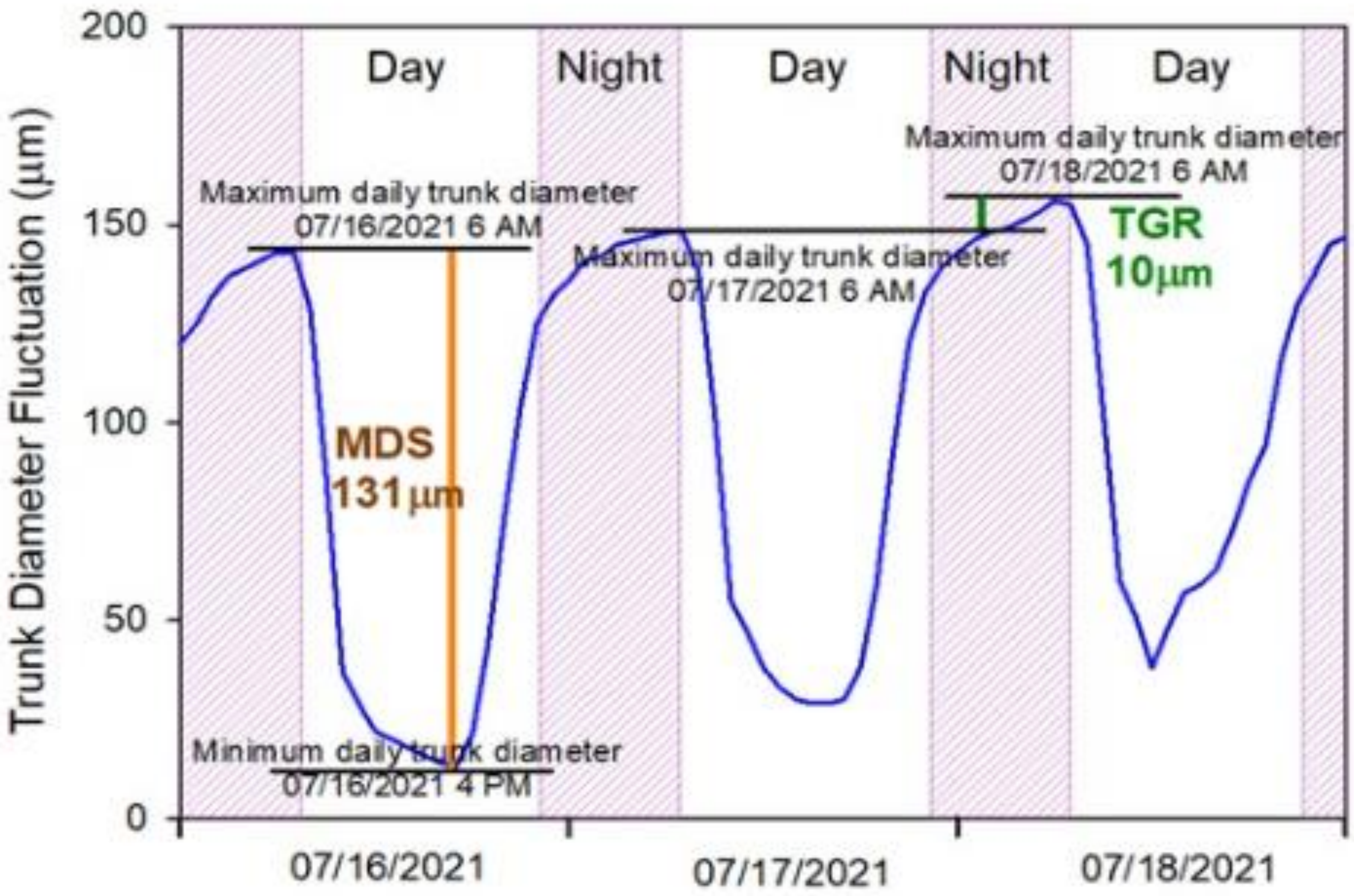
DENDROMETERS

Maximum Daily Shrinkage (MDS)

MDS is calculated as the difference between the max and the min trunk diameter recorded during the same day.

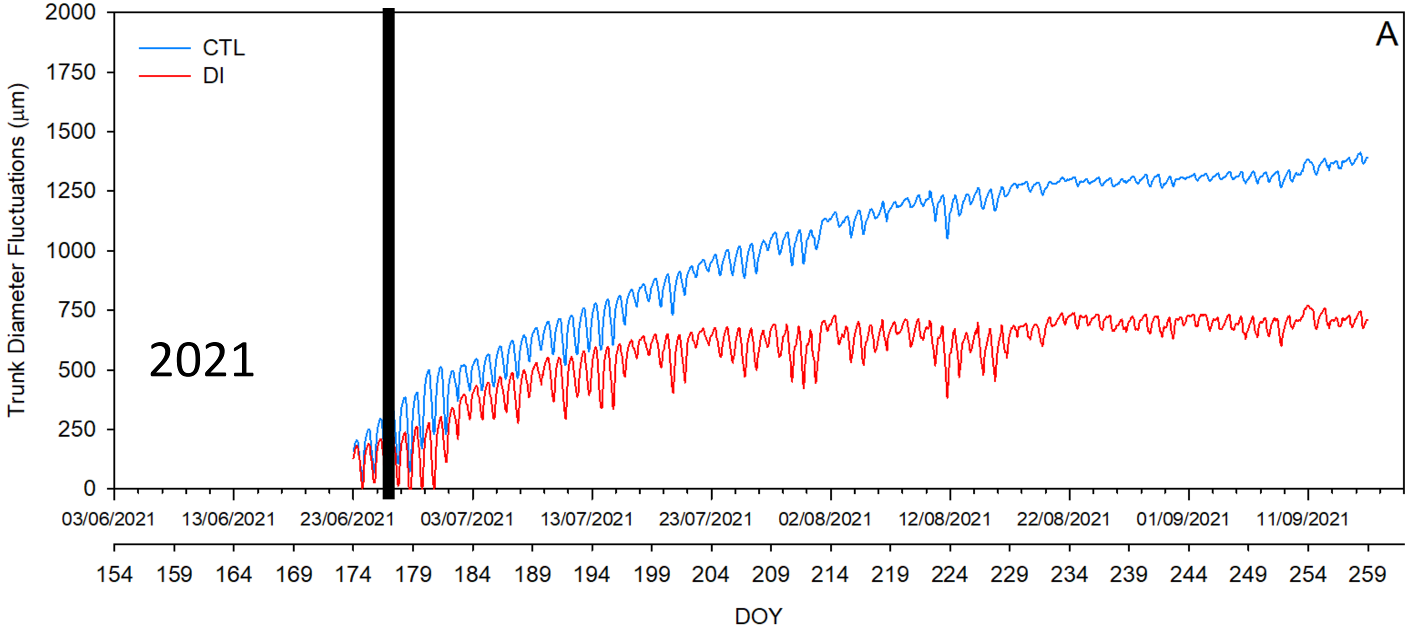
Trunk Growth Rate (TGR)

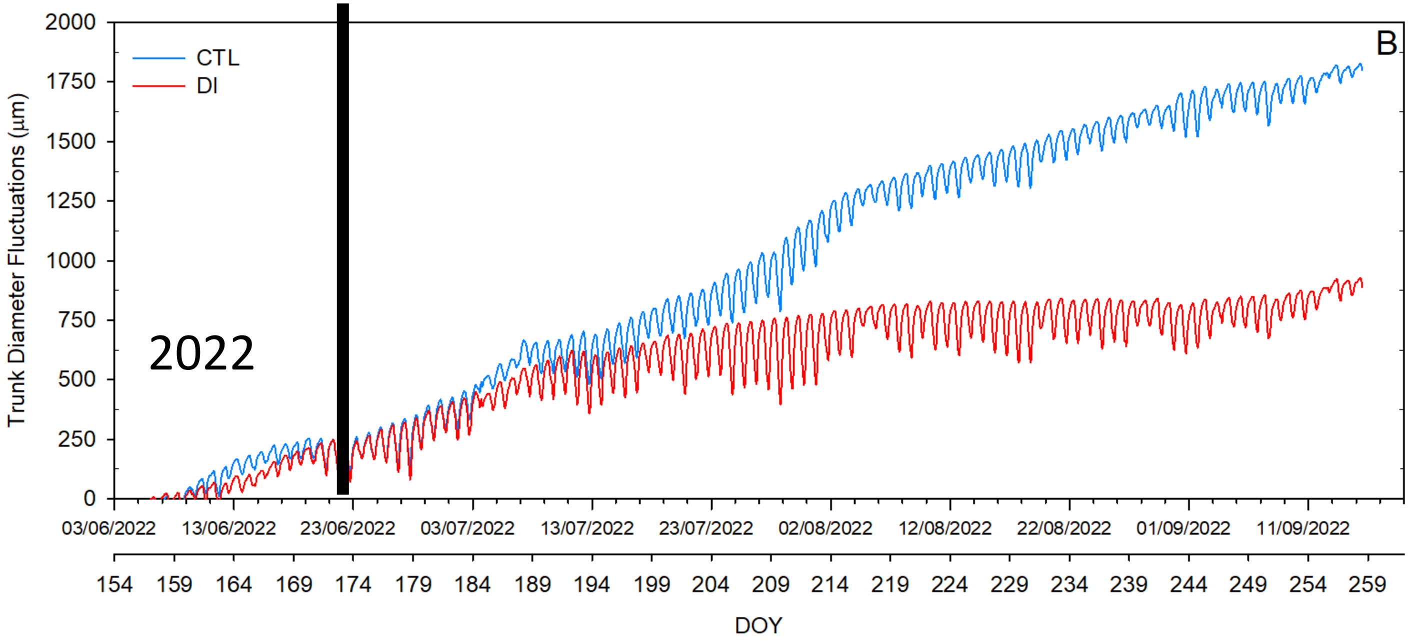
TGR is calculated as the difference between the maximum trunk diameter recorded on two consecutive days.



Results – Seasonal Evolution

TRUNK DIAMETER FLUCTUATIONS



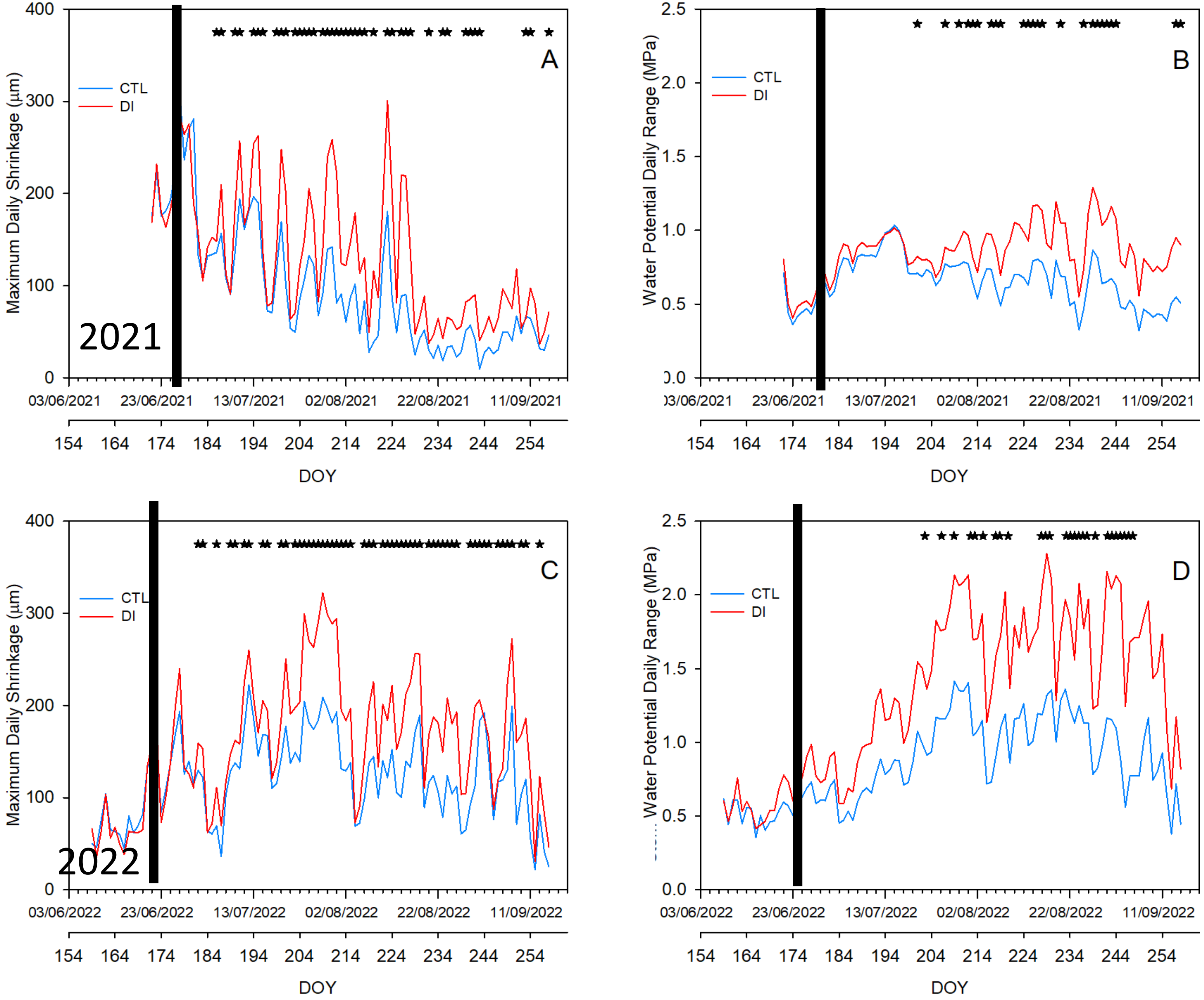


Both years before the irrigation change, the trunk diameter fluctuations were similar.

Trunk growth was greater for CTL trees in 2021 than in 2022.

Results – Seasonal Evolution

TRUNK DIAMETER FLUCTUATIONS



MDS rapidly detected water stress.

MDS had greater variability in 2022 than in 2021 for both treatments.

Ψtrunk daily range was not as sensitive as the midday trunk water potential.

Results – Seasonal Evolution

	R^2	a	b	n
Ψstem vs T daily mean value				
2021	0,68	-0,02	-0,06	24
2022	0,85	-0,03	0,11	24
Ψstem vs T midday				
2021	0,65	-0,02	0,01	24
2022	0,84	-0,03	0,21	24
Ψstem vs VPD daily mean value				
2021	0,69	-0,19	-0,32	24
2022	0,84	-0,16	-0,32	24
Ψstem vs VPD midday				
2021	0,68	-0,15	-0,32	24
2022	0,88	-0,20	-0,3	24
Ψstem vs ET0				
2021	0,41	-0,05	-0,29	24
2022	0,37	-0,05	-0,32	24

Ψstem measured with the Pressure Chamber

24 values per season

The relationship between Ψstem and the climatic variables showed similar coefficients of determination for VPD and T.

Results – Seasonal Evolution

	R^2	a	b	n
Ψtrunk vs T daily mean value				
2021	0,77	-0,03	0,01	101
2022	0,69	-0,04	0,35	101
Ψtrunk vs T midday				
2021	0,84	-0,03	0,12	101
2022	0,75	-0,04	0,46	101
Ψtrunk vs VPD daily mean value				
2021	0,79	-0,23	-0,32	101
2022	0,76	-0,26	-0,17	101
Ψtrunk vs VPD midday				
2021	0,85	-0,20	-0,30	101
2022	0,80	-0,30	-0,16	101
Ψtrunk vs ET0				
2021	0,61	-0,06	-0,29	101
2022	0,34	-0,08	-0,19	101

Ψtrunk measured with the Microtensiometers

101 values per season

The relationship between Ψtrunk and the climatic variables showed similar coefficients of determination for VPD and T.

Results – Seasonal Evolution

	R^2	a	b	n
MDS vs T daily mean value				
2021	0,68	12,72	-230,54	85
2022	0,55	8,01	-74,11	101
MDS vs T midday				
2021	0,68	11,93	-264,95	85
2022	0,64	7,80	-103,16	101
MDS vs VPD daily mean value				
2021	0,67	94,25	-69,61	85
2022	0,61	46,34	19,73	101
MDS vs VPD midday				
2021	0,71	80,50	-76,72	85
2022	0,58	52,15	22,27	101
MDS vs ET0 daily				
2021	0,56	23,33	-76,37	85
2022	0,53	19,43	-15,16	101

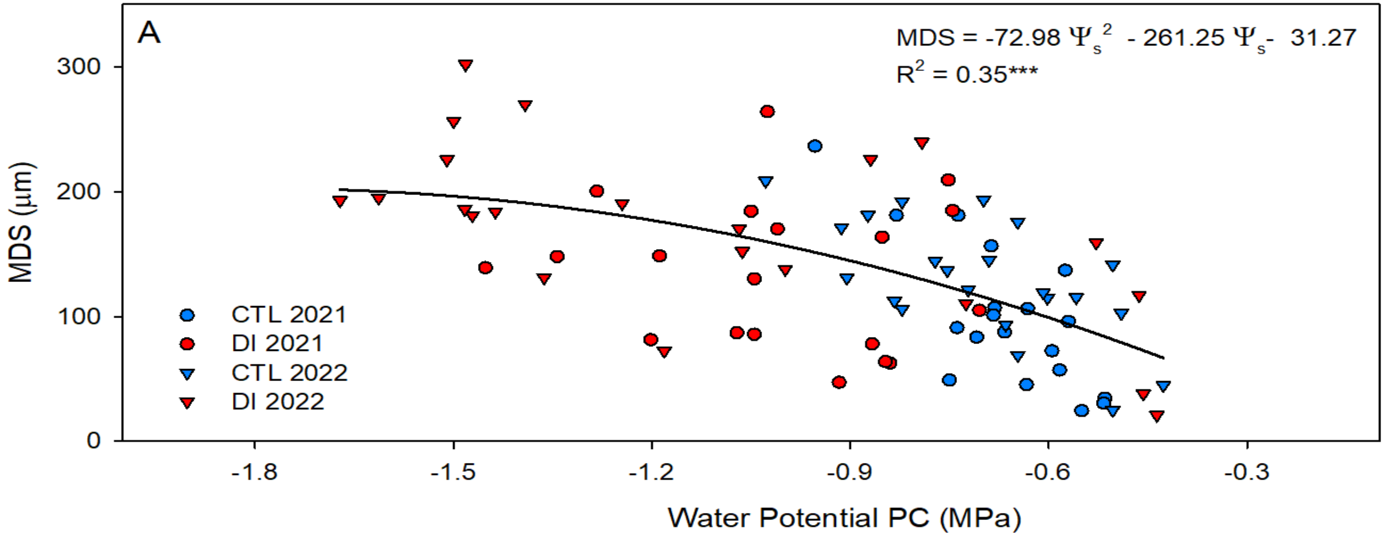
MDS

Between 85 and 101 values per seasons

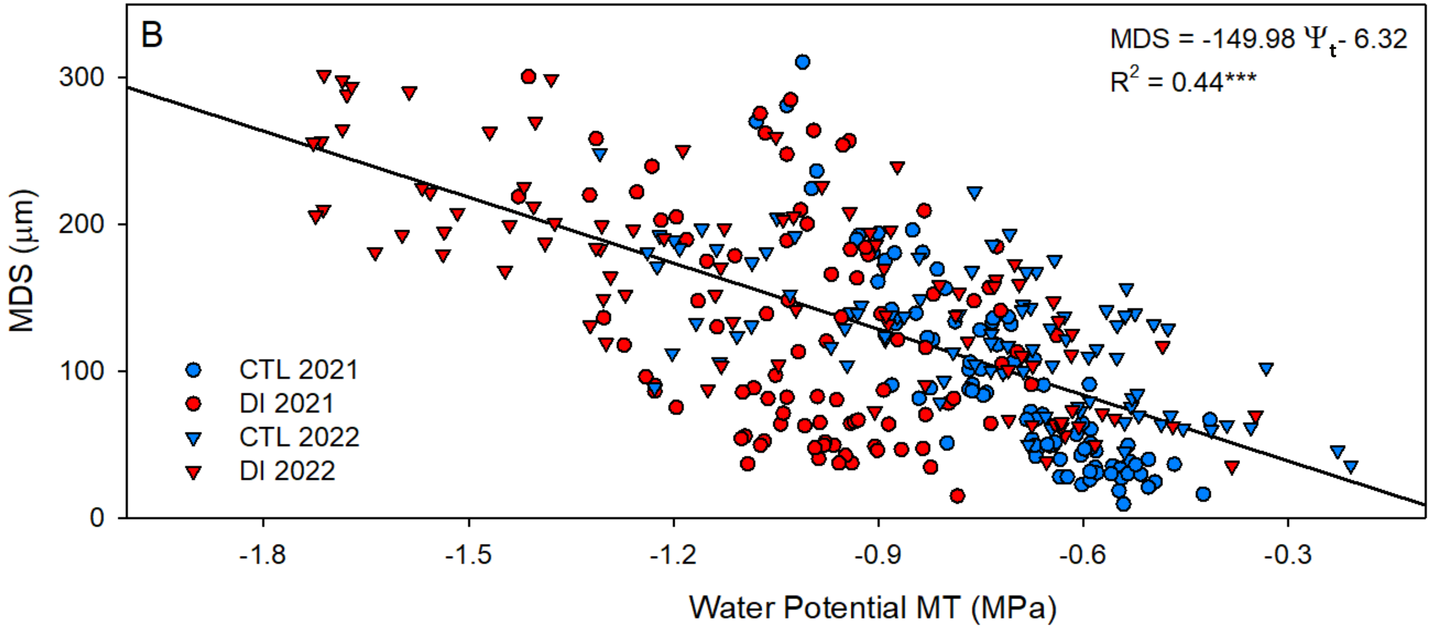
The relationship between MDS and the climatic variables measured at midday showed slightly lower coefficients of determination for VPD and T than Ψstem and Ψtrunk, but higher for ET0.

Results – Seasonal Evolution

STEM WATER POTENTIAL *vs.* MDS



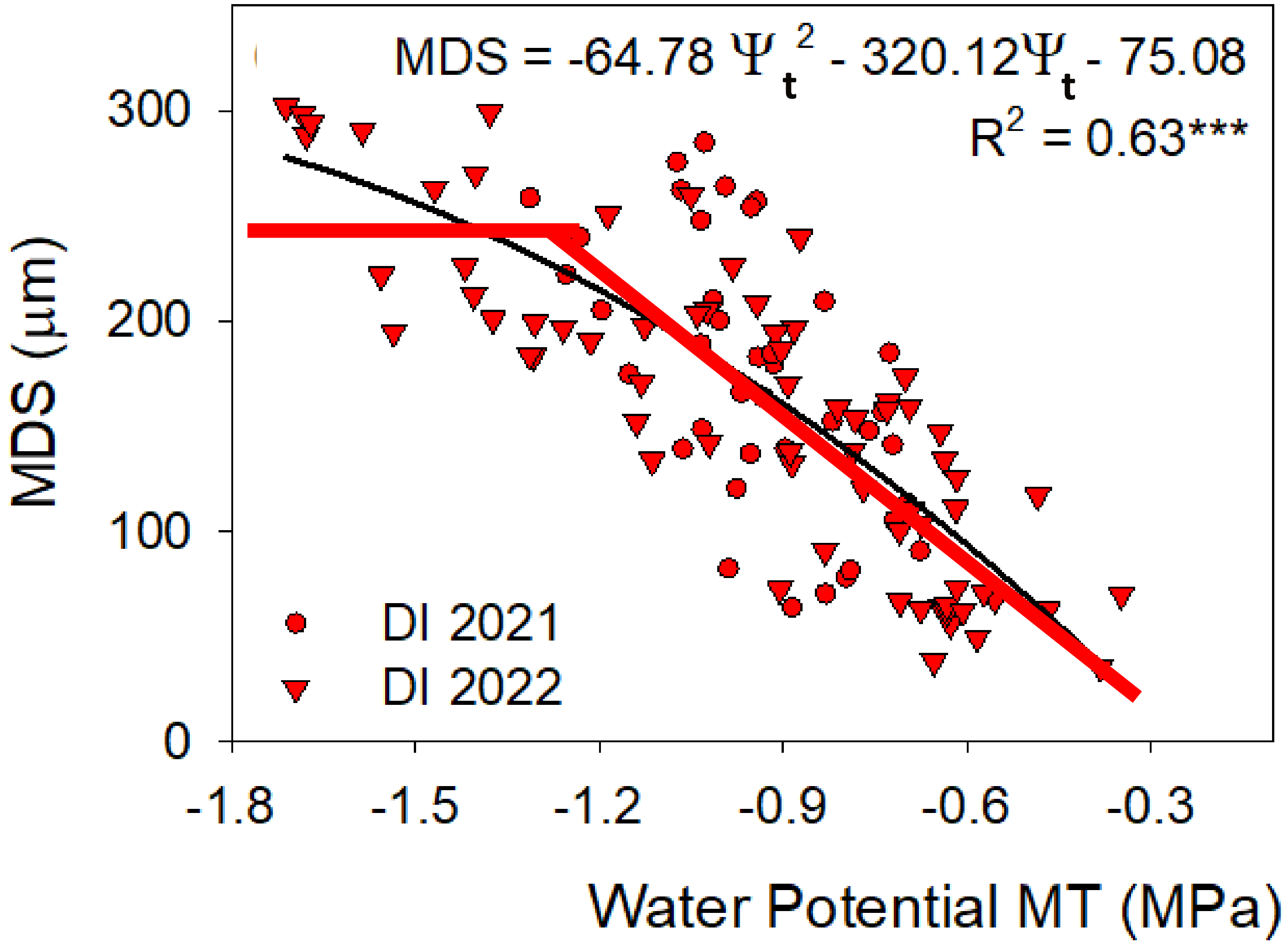
TRUNK WATER POTENTIAL *vs.* MDS



We noticed that a polynomial regression could be obtained with the pressure chamber, while a linear regression was the best fit for the microtensiometers.

Results – Seasonal Evolution

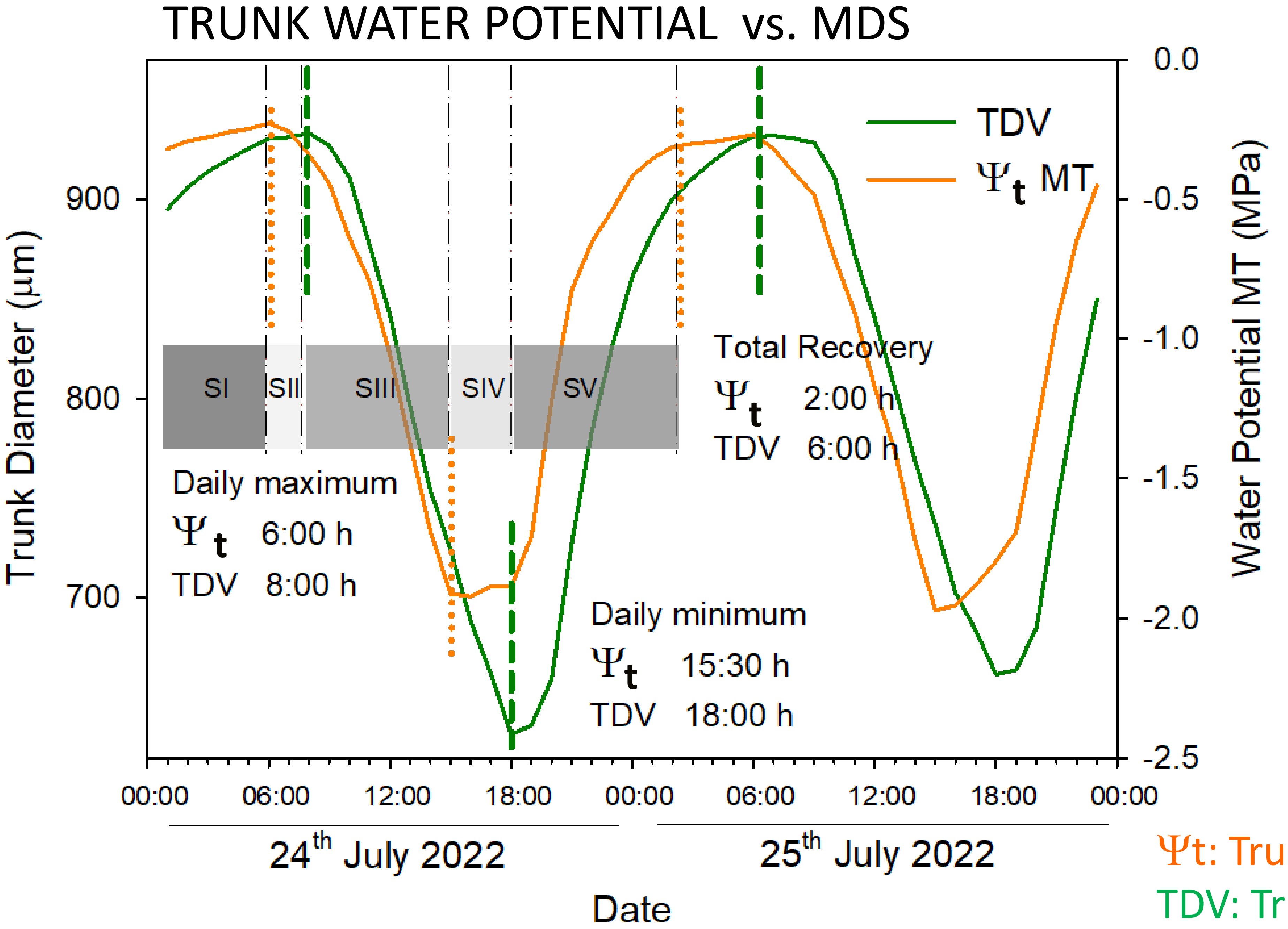
TRUNK WATER POTENTIAL vs. MDS



Midday Ψtrunk measured with the microtensiometers during the month prior and after the deficit irrigation was imposed.

MDS is not sensitive to detect water stress when Ψtrunk is lower than -1.2 MPa.

Results – Seasonal Evolution



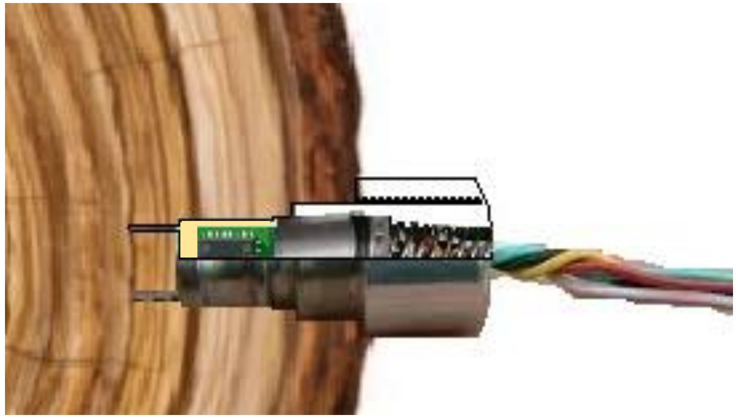
Daily maximum and minimum Ψtrunk values were reached two hours and two hours and a half earlier than the maximum and the minimum trunk diameter were recorded.

Ψt: Trunk Water Potential

TDV: Trunk Diameter Variations

Conclusions – TAKE HOME MESSAGE

<u>Ψtrunk was a sensitive water status indicator.</u>



MT provided continuous and accurate information of the tree water status.

Ψtrunk resulted strongly related to the VPD.

Ψtrunk values measured using MT had greater variability than Ψstem from PC and underestimated water potential values below -1.5 MPa.

THANK YOU FOR YOUR ATTENTION!

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IRTA Institute of Agrifood Research and Technology